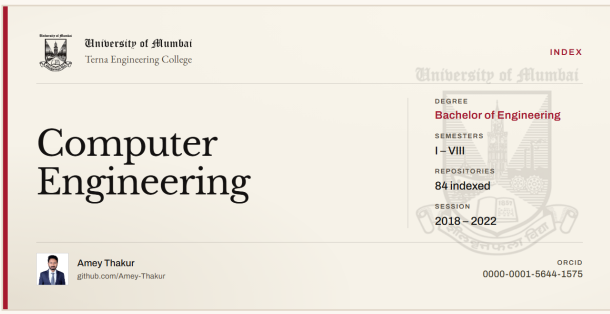
THE COMPLETE ARCHIVE

Computer Engineering

B.E., University of Mumbai

M.Eng., University of Windsor

96	12	16
REPOSITORIES	SEMESTERS	MAJOR PROJECTS



Amey Thakur

github.com/Amey-Thakur

A Note from Amey Thakur

Education is not a destination but a continuous journey. These repositories represent my commitment to that philosophy: a deliberate effort to preserve, organize, and revisit the knowledge acquired across both my undergraduate and graduate studies in Computer Engineering.

The creation of this archive stems from a fundamental belief: **true learning transcends the classroom and extends far beyond graduation**. As I navigate my professional career, I recognize that the concepts, methodologies, and problem-solving frameworks developed during these formative years remain invaluable. However, knowledge, when not actively maintained, fades. These repositories serve as my intellectual foundation: a structured collection I can return to for relearning, reference, and reflection.

Why these repositories exist

Knowledge Preservation	To capture and maintain the depth of understanding developed across twelve semesters of rigorous study.
Continuous Learning	To create a resource that supports lifelong learning, enabling me to revisit fundamental principles as I encounter advanced challenges.
Academic Integrity	To document my authentic academic journey - every concept studied, every project implemented, every problem solved.
Open Contribution	To share these materials with the broader community, believing that knowledge grows when freely exchanged.

These are more than a collection of files or a digital archive. They are a testament to years of intellectual growth, a record of challenges overcome, and a foundation upon which future learning will be built. By making this work publicly available, I hope it serves not only my own continued education but also assists others on their learning journeys.

> **Note**
> All materials in these repositories were created, compiled, and organized by me throughout my undergraduate program (2018-2022) and my graduate program (2023-2024), as part of my coursework, laboratory assignments, and project implementations.

— Amey Thakur

Contents

Start Here

What is in here.

96	12	16	17
REPOSITORIES	SEMESTERS	MAJOR PROJECTS	RECORDINGS

Who it is for

You are on the same syllabus	Every subject is here in the order it was taught, with the notes, the laboratory work and the reports that went with it.
You are teaching yourself	The projects come with working code and, for many of them, a recording of the thing running.
You are assessing my work	This is the whole record, linked and dated, with nothing curated out.

Tip
Everything linked from this document is published under Creative Commons Attribution 4.0. You may read it, copy it, adapt it and build on it, including commercially. Credit me and you are done.

Four ways in

Browse the coursework
Every subject I took, in semester order, each linked to its repository.

Semester sections

Go straight to the projects
The work that went past a weekly assignment, with what each one does.

Major Projects

Watch it running
Recorded demonstrations of the projects and laboratory sessions.

youtube.com/@ameythakur

Find me elsewhere
Publications, code, profiles and everything outside this archive.

amey-thakur.github.io

How to Use This Guide

It is an index, not a textbook	Every row names a repository and links to it. The repository holds the material.
Semester order, not alphabetical	Subjects appear as I took them. A later course usually assumes an earlier one.
Course and laboratory share a repository	Where a subject had both, both codes point at the same repository.
Projects are listed on their own	Substantial work has its own section, because it did not take the same effort as a weekly assignment.
The repository is the live copy	This document is dated. Where the two disagree, the repository is right.

> **Important**
> This is my own archive. It is not a course, a curriculum, an accredited programme, or an official publication of the University of Mumbai, Terna Engineering College, or the University of Windsor. The institutional marks appear only to place the work in context.

My Computer Engineering Archive

Two degrees, twelve semesters, 96 repositories. I keep the two records apart rather than merging them into one list.

	B.E. Computer Engineering	M.Eng. Computer Engineering
University	University of Mumbai	University of Windsor
Institution	Terna Engineering College	—
Structure	Eight semesters	Four semesters
Repositories indexed	82	14
Semesters archived	6	4
Featured projects	14	2

B.E. Computer Engineering

The undergraduate half. Four years at Terna Engineering College, under the University of Mumbai.

Degree	B.E. Computer Engineering
University	University of Mumbai
Institution	Terna Engineering College
Structure	Eight semesters, four academic years
Repositories indexed	82
Semesters with archived work	6

Semester I

First Year Engineering, common to all branches

> **Note**
> My first year predates this archive. I list both semesters so the full eight-semester structure is visible, but I kept no repositories from them.

Code	Subject	Repository
FEC101	Applied Mathematics - I	Not archived
FEC102	Applied Physics - I	Not archived
FEC103	Applied Chemistry - I	Not archived
FEC104	Engineering Mechanics	Not archived
FEC105	Basic Electrical Engineering	Not archived
FEC106	Environmental Studies	Not archived
FEL101	Basic Workshop Practice - I	Not archived

Semester II

First Year Engineering, common to all branches

Code	Subject	Repository
FEC201	Applied Mathematics - II	Not archived
FEC202	Applied Physics - II	Not archived
FEC203	Applied Chemistry - II	Not archived
FEC204	Engineering Drawing	Not archived
FEC205	Structured Programming Approach	Not archived
FEC206	Communication Skills	Not archived
FEL201	Basic Workshop Practice - II	Not archived

Semester III

Code	Subject	Repository
CSC301	Applied Mathematics - III	APPLIED-MATHEMATICS-III Course Question Papers · Reference Books
CSC302 **CSL301**	Digital Logic Design and Analysis Digital System Lab	DIGITAL-LOGIC-DESIGN-AND-ANALYSIS-AND-DIGITAL-SYSTEM-LAB Course and laboratory Question Papers · Reference Books
CSC303	Discrete Mathematics	DISCRETE-MATHEMATICS Course Question Papers · Reference Books
CSC304 **CSL302**	Electronic Circuits and Communication Fundamentals Basic Electronics Lab	ELECTRONIC-CIRCUITS-AND-COMMUNICATION-FUNDAMENTALS-AND-BASIC-ELECTRONICS-LAB Course and laboratory Question Papers · Reference Books
CSC305 **CSL303**	Data Structures Data Structure Lab	DATA-STRUCTURES-AND-DATA-STRUCTURES-LAB Course and laboratory Question Papers · Reference Books · DS · Data Structures Lab
CSL304	OOPM (Java) Lab	OOPM-JAVA-LAB Laboratory Quizzes · Reference Books · OOPM · OOPM Lab · OOPM Mini-Project

Project work

HANGMAN-WORD-GAME	A classic graphical Hangman game implemented using Java Applets and AWT/Swing components

Semester IV

Code	Subject	Repository
CSC401	Applied Mathematics - IV	APPLIED-MATHEMATICS-IV Course Reference Books
CSC402 **CSL401**	Analysis of Algorithm Analysis of Algorithms Lab	ANALYSIS-OF-ALGORITHM-AND-ANALYSIS-OF-ALGORITHM-LAB Course and laboratory Assignments · Reference Books · AOA · AOA Lab
CSC403 **CSL403**	Computer Organization and Architecture Processor Architecture Lab	COMPUTER-ORGANIZATION-AND-ARCHITECTURE-AND-PROCESSOR-ARCHITECTURE-LAB Course and laboratory Assignments · Reference Books · PAL · Processor Architecture Lab
CSC404 **CSL402**	Computer Graphics Computer Graphics Lab	COMPUTER-GRAPHICS-AND-COMPUTER-GRAPHICS-LAB Course and laboratory Assignments · Reference Books · CG · Computer Graphics Lab
CSC405 **CSL404**	Operating System Operating System Lab	OPERATING-SYSTEM-AND-OPERATING-SYSTEM-LAB Course and laboratory Assignments · Reference Books · OS · Operating System Lab
CSL405	Open Source Tech Lab	OPEN-SOURCE-TECH-LAB Laboratory Assignments · Reference Books · Open Source Tech Lab

Project work

AR-STACK-GAME	An augmented reality stack-builder game for Android developed using Google's ARCore, Unity, and C#
SIMPLE-AND-COMPOUND-INTEREST-CALCULATOR	A Shell Script program designed to calculate simple and compound interest using user-provided inputs
COVID19-WEB-SCRAPER	Scraping and visualizing India's real-time COVID-19 data from the Ministry of Health and Family Welfare (MoHFW) dataset

Semester V

Code	Subject	Repository
CSC501 **CSL501**	Microprocessor Microprocessor Lab	MICROPROCESSOR-AND-MICROPROCESSOR-LAB Course and laboratory Assignments · Classwork · Question Papers · Semester Exam · Online Exam · Quizzes · and 3 more
CSC502 **CSL503**	Database Management System Database Management System Lab	DATABASE-MANAGEMENT-SYSTEM-AND-DATABASE-MANAGEMENT-SYSTEM-LAB Course and laboratory Assignments · MEGA NOTES · Question Papers · Semester Exam · Online Exam · Quizzes · and 5 more
CSC503 **CSL502**	Computer Network Computer Network Lab	COMPUTER-NETWORK-AND-COMPUTER-NETWORK-LAB Course and laboratory Assignments · Question Papers · Semester Exam · Practical Exam · Online Exam · Submission Report · and 3 more
CSC504	Theory of Computer Science	THEORY-OF-COMPUTER-SCIENCE Course Assignments · Question Papers · Semester Exam · Online Exam · Quizzes · Reference Books · Syllabus
CSDLO5011	Multimedia System	MULTIMEDIA-SYSTEM Course Semester Exam · Online Exam · Quizzes · Reference Books · Syllabus
CSL504	Web Design Lab	WEB-DESIGNING-LAB Laboratory Online Exam · Quizzes · Submission Report · Reference Books · Syllabus · WDL · and 3 more
CSL505	Business Communication & Ethics	BUSINESS-COMMUNICATION-AND-ETHICS Course Assignments · Quizzes · Reference Books · Syllabus · Activities · Google Meet (dee-qebi-amh)

Project work

8086-ASSEMBLY-LANGUAGE-PROGRAMS	525 documented Intel 8086 assembly programs, with a browser simulator that assembles and runs every one of them: full instruction set, nine flags, macros, DOS and BIOS services, and step-by-step debugging. Verified by 2010 tests
CAR-RENTAL-SYSTEM	A web-based Car Rental Database Management System designed for online vehicle booking and administrative management, featuring separate user and admin interfaces
CHAT-ROOM	A web-based real-time chat application developed using HTML, CSS, JavaScript, PHP, AJAX, and MySQL, demonstrating asynchronous communication and dynamic Document Object Model (DOM) updates

Semester VI

Code	Subject	Repository
CSC601 **CSL601**	Software Engineering Software Engineering Lab	SOFTWARE-ENGINEERING-AND-SOFTWARE-ENGINEERING-LAB Course and laboratory Assignments · Question Papers · Semester Exam · Quizzes · Submission Report · Reference Books · and 3 more
CSC602 **CSL602**	System Programming and Compiler Construction System Software Lab	SYSTEM-PROGRAMMING-AND-COMPILER-CONSTRUCTION-AND-SYSTEM-SOFTWARE-LAB Course and laboratory Assignments · Classwork · Question Papers · Semester Exam · Quizzes · Submission Report · and 4 more
CSC603 **CSL603**	Data Warehousing and Mining Data Warehousing and Mining Lab	DATA-WAREHOUSING-AND-MINING-AND-DATA-WAREHOUSING-AND-MINING-LAB Course and laboratory Assignments · Question Papers · Semester Exam · Reference Books · Syllabus · DWM Lab · Internal Assessment Test
CSC604 **CSL604**	Cryptography and System Security System Security Lab	CRYPTOGRAPHY-AND-SYSTEM-SECURITY-AND-SYSTEM-SECURITY-LAB Course and laboratory Assignments · Question Papers · Semester Exam · Quizzes · Submission Report · Reference Books · and 5 more
CSDLO6021	Machine Learning	MACHINE-LEARNING Course Assignments · Question Papers · Semester Exam · Quizzes · Submission Report · Reference Books · and 2 more

Project work

DIGITAL-BOOKSTORE	A comprehensive web-based e-commerce platform designed to support book discovery, secure user authentication, and persistent shopping cart management
WHITE-BOX-CARTOONIZATION	AI-powered web app that transforms photos into cartoon-style images using deep learning (GAN + U-Net architecture)

Semester VII

Code	Subject	Repository
CSC701 **CSL701**	Digital Signal and Image Processing Digital Signal and Image Processing Lab	DIGITAL-SIGNAL-AND-IMAGE-PROCESSING-AND-DIGITAL-SIGNAL-AND-IMAGE-PROCESSING-LAB Course and laboratory Assignments · Question Papers · Semester Exam · Reference Books · Syllabus · DSIP Lab · Internal Assessment Test
CSC702 **CSL702**	Mobile Communication and Computing Mobile Application Development Lab	MOBILE-COMMUNICATION-AND-COMPUTING-AND-MOBILE-APPLICATION-DEVELOPMENT-LAB Course and laboratory Assignments · Question Papers · Semester Exam · Quizzes · Reference Books · Syllabus · and 3 more
CSC703 **CSL703**	Artificial Intelligence and Soft Computing Artificial Intelligence and Soft Computing Lab	ARTIFICIAL-INTELLIGENCE-AND-SOFT-COMPUTING-AND-ARTIFICIAL-INTELLIGENCE-AND-SOFT-COMPUTING-LAB Course and laboratory Assignments · Question Papers · Semester Exam · Quizzes · Submission Report · Reference Books · and 3 more
CSDLO7032 **CSL704**	Big Data Analytics Computational Lab - I	BIG-DATA-ANALYTICS-AND-COMPUTATIONAL-LAB-I Course and laboratory Assignments · Classwork · Mini-Project · Question Papers · Semester Exam · Quizzes · and 6 more
ILO7013	Management Information System	MANAGEMENT-INFORMATION-SYSTEM Course Question Papers · Semester Exam · Reference Books · Syllabus · Internal Assessment Test

Project work

OPTIMIZING-STOCK-TRADING-STRATEGY-WITH-K-MEANS-CLUSTERING	Big Data Analytics [BDA] Mini Project
QUADTREE-VISUALIZER	A high-performance interactive simulation visualizing the efficiency of the QuadTree data structure in spatial partitioning and collision detection, built with Next.js and HTML5 Canvas

Semester VIII

Code	Subject	Repository
CSC801 **CSL801**	Human Machine Interaction Human Machine Interaction Lab	HUMAN-MACHINE-INTERACTION-AND-HUMAN-MACHINE-INTERACTION-LAB Course and laboratory Assignments · MEGA NOTES · Semester Exam · Internal Assessment Tests · Quizzes · Submission Report · and 4 more
CSC802 **CSL802**	Distributed Computing Distributed Computing Lab	DISTRIBUTED-COMPUTING-AND-DISTRIBUTED-COMPUTING-LAB Course and laboratory Assignments · Semester Exam · Internal Assessment Tests · Quizzes · Submission Report · Reference Books · and 2 more
DLO8012 **CSL804**	Natural Language Processing Computational Lab - II	NATURAL-LANGUAGE-PROCESSING-AND-COMPUTATIONAL-LAB-II Course and laboratory Assignments · Semester Exam · Internal Assessment Tests · Submission Report · Reference Books · Computational Lab - II
ILO8022	Finance Management	FINANCE-MANAGEMENT Course MEGA NOTES · Semester Exam · Internal Assessment Tests · Submission Report · Reference Books · Zerodha Varsity - Karthik Rangappa
CSL803	Cloud Computing Lab	CLOUD-COMPUTING-LAB Laboratory Assignments · Quizzes · Submission Report · Reference Books · CCL · CCL Experiments · CCL Mini-Project

Project work

ONLINE-CHESS-GAME	A real-time multiplayer chess application featuring instant move synchronization, in-game chat, and AI integration, designed with a focus on human-computer interaction principles
AWS-CERTIFIED-CLOUD-PRACTITIONER-CLF-C01	AWS CERTIFIED CLOUD PRACTITIONER (AWS CLF-C01)
TEXT-SUMMARIZER	Machine Learning Project to Compare and Evaluate Text Summarization Algorithms Using SpaCy, NLTK, Gensim, and Sumy
QUADTREE-VISUALIZER	A high-performance interactive simulation visualizing the efficiency of the QuadTree data structure in spatial partitioning and collision detection, built with Next.js and HTML5 Canvas

B.E. Major Projects

Work that went past a weekly assignment. Each entry gives the subject and semester it came from.

Quadtree Visualizer

Major Capstone Project (Semesters VII-VIII)

A high-performance interactive simulation visualizing the efficiency of the QuadTree data structure in spatial partitioning and collision detection, built with Next.js and HTML5 Canvas

QUADTREE-VISUALIZER

AWS Certified Cloud Practitioner

Cloud Computing Certification (Semester VIII)

AWS CERTIFIED CLOUD PRACTITIONER (AWS CLF-C01)

AWS-CERTIFIED-CLOUD-PRACTITIONER-CLF-C01

Text Summarizer

Natural Language Processing (Semester VIII)

Machine Learning Project to Compare and Evaluate Text Summarization Algorithms Using SpaCy, NLTK, Gensim, and Sumy

TEXT-SUMMARIZER

Online Chess Game

Human-Machine Interaction (Semester VIII)

A real-time multiplayer chess application featuring instant move synchronization, in-game chat, and AI integration, designed with a focus on human-computer interaction principles

ONLINE-CHESS-GAME

K-Means Stock Trading

Big Data Analytics (Semester VII)

Big Data Analytics [BDA] Mini Project

OPTIMIZING-STOCK-TRADING-STRATEGY-WITH-K-MEANS-CLUSTERING

White-Box Cartoonization

Computer Vision Mini-Project (Semester VI)

AI-powered web app that transforms photos into cartoon-style images using deep learning (GAN + U-Net architecture)

WHITE-BOX-CARTOONIZATION

Digital Bookstore

Software Engineering (Semester VI)

A comprehensive web-based e-commerce platform designed to support book discovery, secure user authentication, and persistent shopping cart management

DIGITAL-BOOKSTORE

Car Rental System

Database Management (Semester V)

A web-based Car Rental Database Management System designed for online vehicle booking and administrative management, featuring separate user and admin interfaces

CAR-RENTAL-SYSTEM

Chat Room

Web Designing (Semester V)

A web-based real-time chat application developed using HTML, CSS, JavaScript, PHP, AJAX, and MySQL, demonstrating asynchronous communication and dynamic Document Object Model (DOM) updates

CHAT-ROOM

8086 ASM

Microprocessor Programming (Semester V)

525 documented Intel 8086 assembly programs, with a browser simulator that assembles and runs every one of them: full instruction set, nine flags, macros, DOS and BIOS services, and step-by-step debugging. Verified by 2010 tests

8086-ASSEMBLY-LANGUAGE-PROGRAMS

COVID-19 Web Scraper

Open Source Technologies (Semester IV)

Scraping and visualizing India's real-time COVID-19 data from the Ministry of Health and Family Welfare (MoHFW) dataset

COVID19-WEB-SCRAPER

Interest Calculator

Operating Systems (Semester IV)

A Shell Script program designed to calculate simple and compound interest using user-provided inputs

SIMPLE-AND-COMPOUND-INTEREST-CALCULATOR

AR Stack Game

Computer Graphics (Semester IV)

An augmented reality stack-builder game for Android developed using Google's ARCore, Unity, and C#

AR-STACK-GAME

Hangman Word Game

Object Oriented Programming (Semester III)

A classic graphical Hangman game implemented using Java Applets and AWT/Swing components

HANGMAN-WORD-GAME

Selected Technical Work

Work I did alongside the curriculum, not for it.

> **Tip**
> None of this was assigned. Several of these started as challenges I set myself.

Artificial Intelligence & Neural Research

ResearchGate Archive	A technical companion repository archiving code implementations and research artifacts shared on ResearchGate
Deepfake Audio Synthesis	Deepfake Audio, A neural voice cloning studio powered by SV2TTS technology
Sentiment Analyzer	An NLP engine utilizing rule-based linguistic patterns for precise sentiment quantification
Depression Detection	Machine Learning Project for Depression Detection Using Tweets

Machine Learning & Predictive Analytics

Stock Trading Strategy RL	Machine Learning Project to Optimize Stock Trading Strategies Using Reinforcement Learning
Reinforcement Learning Strategy	Machine Learning Project to Optimize Stock Trading Strategies Using Reinforcement Learning
Kaggle Courses	A comprehensive archive of completed Kaggle Data Science, Machine Learning, and Data Manipulation courses, documenting structured Python exercises and certifications
House Price Prediction	Machine Learning Project to Predict House Prices in Bangalore

Professional Machine Learning Missions (TSF)

Unsupervised ML	Task: From the given 'Iris' dataset, predict the optimum number of clusters and represent it visually
Supervised ML	Task: To predict the percentage of a student based on the number of study hours

Full Stack Engineering & Distributed Computing

React Todo App	A Todo application developed using React
Tic-Tac-Toe in Angular	A Tic-Tac-Toe game application developed using the Angular framework
JavaScript Frameworks	Comparative Implementation of Todo Applications Across JavaScript Frameworks
Hangman in Django	A Hangman game web application developed using Django
Hadoop	HADOOP - Matrix Multiplication

Ruby Language & Algorithmic Foundations

30 Days of Ruby	30-Day Ruby Programming Challenge by Amey Thakur and Mega Satish, focused on mastering Ruby fundamentals, object-oriented programming, Ruby on Rails development, and full-stack web engineering with cloud deployment
Hangman in Ruby	A Ruby terminal-based Hangman game showcasing object-oriented design, modular architecture, and dynamic game state management
Tic-Tac-Toe in Ruby	A Ruby terminal-based Tic Tac Toe game showcasing object-oriented programming (OOP), modular design, and structured game state management
Ruby on Rails FriendsApp	Friends App Using Ruby on Rails with PostgreSQL database deployed on Heroku. [Heroku Render]
RailsFriends	A Ruby on Rails Friends List application showcasing MVC architecture, RESTful CRUD operations, Active Record ORM, and SQLite3 database integration

Polyglot Mastery & Logic Synthesis

Python Shorts	A curated collection of 100+ Python programs, algorithms, and data structure implementations for problem solving and computational practice
Python Crash Course	IIT Ropar, Diginique Techlabs: Python Crash Course in Data Science, Machine Learning, and Artificial Intelligence
30 Days of R	30-Day R Programming Challenge by Amey Thakur and Mega Satish, focused on statistical computing, data visualization, and applied data science
50 Days of Julia	50-Day Julia Programming Challenge by Amey Thakur and Mega Satish, focused on computational modeling, high-performance computing, and machine learning

HMI Lab & Neural Game Development

The Math Game	A simple, interactive multiplication game designed to test and improve arithmetic skills through rapid-fire problem solving
Math Sprint Game	A fast-paced educational web game where players race against time to solve math equations, testing speed and accuracy
Rock Paper Scissors	A classic Rock Paper Scissors game implemented as a responsive web application, featuring interactive gameplay and score tracking
ATVM Interface	A simulation of the Automatic Ticket Vending Machine (ATVM) interface found at railway stations, designed with Human-Machine Interaction (HMI) principles
Flappy Bird	A Python + Pygame recreation of Flappy Bird, engineered with precise collision physics, optimized game loop timing, and arcade-style responsiveness
Pong Game	A modern Python + Pygame reconstruction of the original 1972 Pong, built with accurate collision physics and performance-focused game loops

Mobile Application Engineering (Android)

Android Studio Flashlight	A simple flashlight Android application developed in Java using Android Studio
Android Studio Calculator	A simple calculator Android application built using Java in Android Studio

Creative Web Design & Interface Discovery

Search Space Explore Extent	It is a simple Web Design in HTML and CSS. [My first Web Design]
Lunar Design Studio	Lunar Design Studio is a professionally designed architecture portfolio project developed for Architect Mugdha Thakur

Scholarly Achievements & Professional Records

Achievements	Every certification, course completion, badge and award earned by Amey Thakur, each shown as the certificate itself and linked to its original file, with issuer verification where it exists. Anthropic, Google, Microsoft, IBM, Intel, NVIDIA, Apple, Kaggle, Coursera, LinkedIn Learning, Stanford, Harvard and Cambridge, plus research and coursework

M.Eng. Computer Engineering

The graduate half. Four semesters at the University of Windsor: fewer subjects, further into each one.

Degree	M.Eng. Computer Engineering
University	University of Windsor
Structure	Four semesters
Repositories indexed	14
Semesters with archived work	4

Semester I

Code	Subject	Repository
GENG 8000	Engineering Technical Communications	ENGINEERING-TECHNICAL-COMMUNICATIONS Course Assignments · Weekly material, 10 weeks · Reference Books · Discussion Posts · Office Hours
GENG 8010	Engineering Mathematics	ENGINEERING-MATHEMATICS Course Assignments · Weekly material, 11 weeks · Final Exam · Midterm Exams · In-Class Quizzes · Reference Books · WileyPLUS

Semester II

Code	Subject	Repository
GENG 8020	Engineering Project Management	ENGINEERING-PROJECT-MANAGEMENT Course Assignments · Notes · Reference Books · Lecture Slides
GENG 8030	Computational Methods and Modeling	COMPUTATIONAL-METHODS-AND-MODELING-FOR-ENGINEERING-APPLICATIONS Course Matlab Project · Announcements · MATLAB Tutorials and Solutions

Project work

ADAPTIVE-CRUISE-CONTROL	Adaptive Cruise Control (ACC) with Arduino, MATLAB, and Simulink · Real-time speed regulation and proximity-based safety system

Semester III

Code	Subject	Repository
ELEC 8560	Computer Networks	COMPUTER-NETWORKS Course Assignments · Homework · Lecture Notes · Reference Books
ELEC 8900	Machine Learning	MACHINE--LEARNING Course Assignments · ML Project · Weekly material, 9 weeks · In-Class Presentation · Reference Books · Course Project Instructions · datacamp
VIP-CSL	VIP - Community Service Learning	VIP-CSL-FALL-2023 Course Content Modules · Manual & Contract · Submissions · Timeline

Project work

ZERO-SHOT-VIDEO-GENERATION	Zero-Shot Video Generation, A neural video synthesis studio powered by latent diffusion and cross-frame attention

Semester IV

Code	Subject	Repository
ELEC 8330	Computational Intelligence	COMPUTATIONAL-INTELLIGENCE Course Assignments · Final Exam · Reference Books · EAB · MATLAB · Midterm Exam
ELEC 8900	Digital Communications	DIGITAL-COMMUNICATIONS Course Homework · Lecture Notes · Notes · Reference Books · Project · Tutorials

M.Eng. Major Projects

Work that went past a weekly assignment. Each entry gives the subject and semester it came from.

Zero-Shot Video Generation

Machine Learning & Computer Vision (Semester III)

Zero-Shot Video Generation, A neural video synthesis studio powered by latent diffusion and cross-frame attention

ZERO-SHOT-VIDEO-GENERATION

Adaptive Cruise Control

Computational Modeling & Control Systems (Semester II)

Adaptive Cruise Control (ACC) with Arduino, MATLAB, and Simulink · Real-time speed regulation and proximity-based safety system

ADAPTIVE-CRUISE-CONTROL

Student support resources

Services I used as an international student. Part of the degree in practice, and hard to find otherwise.

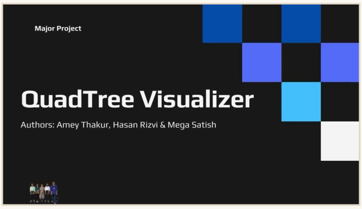
International Student Centre	International Student Centre (ISC), University of Windsor · Orientation Resources & Student Support Hub
Engineering International Student Advising	Engineering International Student Advising (ENG-ISA), University of Windsor · Academic, immigration, and student success resources for international engineering graduate students
Writing Support	Writing Support Desk (WSD), University of Windsor · Academic writing, citation standards (APA, MLA, IEEE, Chicago), and academic integrity resources

Video Demonstrations

I recorded these myself while doing the work. They show it being performed, not described.

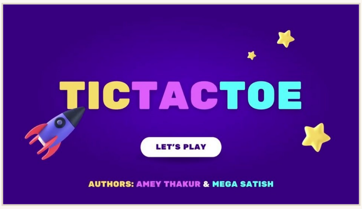
> **Tip**
> Every thumbnail below is a link. 17 videos, 1 hour 25 minutes in total, on my own channel.

Full playlist	youtube.com/playlist?list=PLGOc13Pt03SZ9INe4gyxoZnA4zAXssiLn
Channel	youtube.com/@ameythakur



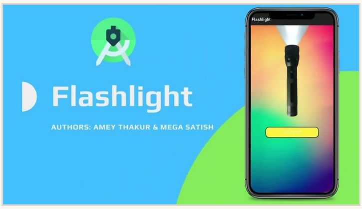
QuadTree Visualizer
4:58

Zero-Shot Text to Video Generation
3:57



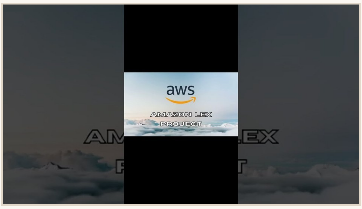
Tic-Tac-Toe Game in Angular Framework
4:06

Calculator App using Android Studio
4:59



Flashlight App using Android Studio
2:04

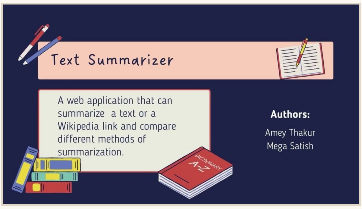
Online Chess Game
1:27



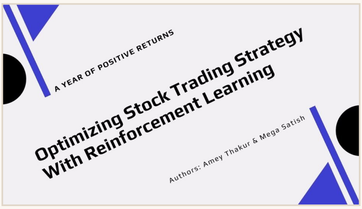
Pizza Ordering Chatbot using Amazon Lex - AWS Project Demo
1:53

Pizza Ordering Chatbot using Amazon Lex - Presentation
1:11

Pizza Ordering Chatbot using Amazon Lex - AWS Project
42:15

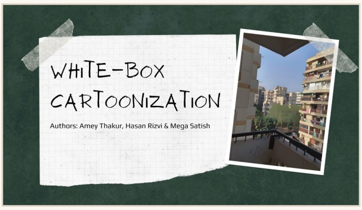


Text Summarizer
1:08



Optimizing Stock Trading Strategy with Reinforcement Learning
1:24

Bangalore House Price Prediction
1:33



White-Box Cartoonization
3:03

DeepFake Audio
3:30

Digital Bookstore
4:40

Chat Room using HTML, PHP, CSS, JavaScript, AJAX
2:29

Hangman Word Game using Applet in Java
1:14

Complete Directory: B.E.

Every repository in this document, A to Z. Each name is a link.

Repository	Subject or project	Where it sits
8086-ASSEMBLY-LANGUAGE-PROGRAMS	525 documented Intel 8086 assembly programs, with a browser simulator that assembles and runs every one of them: full instruction set, nine flags, macros, DOS and BIOS services, and step-by-step debugging. Verified by 2010 tests	Semester V · Project
ACHIEVEMENTS	Every certification, course completion, badge and award earned by Amey Thakur, each shown as the certificate itself and linked to its original file, with issuer verification where it exists. Anthropic, Google, Microsoft, IBM, Intel, NVIDIA, Apple, Kaggle, Coursera, LinkedIn Learning, Stanford, Harvard and Cambridge, plus research and coursework	Technical work · Scholarly Achievements & Professional Records
ANALYSIS-OF-ALGORITHM-AND-ANALYSIS-OF-ALGORITHM-LAB	Analysis of Algorithm · Analysis of Algorithms Lab	Semester IV · CSC402, CSL401
ANDROID-STUDIO-CALCULATOR	A simple calculator Android application built using Java in Android Studio	Technical work · Mobile Application Engineering (Android)
ANDROID-STUDIO-FLASHLIGHT	A simple flashlight Android application developed in Java using Android Studio	Technical work · Mobile Application Engineering (Android)
APPLIED-MATHEMATICS-III	Applied Mathematics - III	Semester III · CSC301
APPLIED-MATHEMATICS-IV	Applied Mathematics - IV	Semester IV · CSC401
AR-STACK-GAME	An augmented reality stack-builder game for Android developed using Google's ARCore, Unity, and C#	Semester IV · Project
ARTIFICIAL-INTELLIGENCE-AND-SOFT-COMPUTING-AND-ARTIFICIAL-INTELLIGENCE-AND-SOFT-COMPUTING-LAB	Artificial Intelligence and Soft Computing · Artificial Intelligence and Soft Computing Lab	Semester VII · CSC703, CSL703

ATVM-INTERFACE	A simulation of the Automatic Ticket Vending Machine (ATVM) interface found at railway stations, designed with Human-Machine Interaction (HMI) principles	Technical work · HMI Lab & Neural Game Development
AWS-CERTIFIED-CLOUD-PRACTITIONER-CLF-C01	AWS CERTIFIED CLOUD PRACTITIONER (AWS CLF-C01)	Semester VIII · Project
BANGALORE-HOUSE-PRICE-PREDICTION	Machine Learning Project to Predict House Prices in Bangalore	Technical work · Machine Learning & Predictive Analytics
BIG-DATA-ANALYTICS-AND-COMPUTATIONAL-LAB-I	Big Data Analytics · Computational Lab - I	Semester VII · CSDLO7032, CSL704
BUSINESS-COMMUNICATION-AND-ETHICS	Business Communication & Ethics	Semester V · CSL505
CAR-RENTAL-SYSTEM	A web-based Car Rental Database Management System designed for online vehicle booking and administrative management, featuring separate user and admin interfaces	Semester V · Project
CHAT-ROOM	A web-based real-time chat application developed using HTML, CSS, JavaScript, PHP, AJAX, and MySQL, demonstrating asynchronous communication and dynamic Document Object Model (DOM) updates	Semester V · Project
CLOUD-COMPUTING-LAB	Cloud Computing Lab	Semester VIII · CSL803
COMPUTER-GRAPHICS-AND-COMPUTER-GRAPHICS-LAB	Computer Graphics · Computer Graphics Lab	Semester IV · CSC404, CSL402
COMPUTER-NETWORK-AND-COMPUTER-NETWORK-LAB	Computer Network · Computer Network Lab	Semester V · CSC503, CSL502
COMPUTER-ORGANIZATION-AND-ARCHITECTURE-AND-PROCESSOR-ARCHITECTURE-LAB	Computer Organization and Architecture · Processor Architecture Lab	Semester IV · CSC403, CSL403
COVID19-WEB-SCRAPER	Scraping and visualizing India's real-time COVID-19 data from the Ministry of Health and Family Welfare (MoHFW) dataset	Semester IV · Project
CRYPTOGRAPHY-AND-SYSTEM-SECURITY-AND-SYSTEM-SECURITY-LAB	Cryptography and System Security · System Security Lab	Semester VI · CSC604, CSL604

DATA-STRUCTURES-AND-DATA-STRUCTURES-LAB	Data Structures · Data Structure Lab	Semester III · CSC305, CSL303
DATA-WAREHOUSING-AND-MINING-AND-DATA-WAREHOUSING-AND-MINING-LAB	Data Warehousing and Mining · Data Warehousing and Mining Lab	Semester VI · CSC603, CSL603
DATABASE-MANAGEMENT-SYSTEM-AND-DATABASE-MANAGEMENT-SYSTEM-LAB	Database Management System · Database Management System Lab	Semester V · CSC502, CSL503
DEEPFAKE-AUDIO	Deepfake Audio, A neural voice cloning studio powered by SV2TTS technology	Technical work · Artificial Intelligence & Neural Research
DEPRESSION-DETECTION-USING-TWEETS	Machine Learning Project for Depression Detection Using Tweets	Technical work · Artificial Intelligence & Neural Research
DIGITAL-BOOKSTORE	A comprehensive web-based e-commerce platform designed to support book discovery, secure user authentication, and persistent shopping cart management	Semester VI · Project
DIGITAL-LOGIC-DESIGN-AND-ANALYSIS-AND-DIGITAL-SYSTEM-LAB	Digital Logic Design and Analysis · Digital System Lab	Semester III · CSC302, CSL301
DIGITAL-SIGNAL-AND-IMAGE-PROCESSING-AND-DIGITAL-SIGNAL-AND-IMAGE-PROCESSING-LAB	Digital Signal and Image Processing · Digital Signal and Image Processing Lab	Semester VII · CSC701, CSL701
DISCRETE-MATHEMATICS	Discrete Mathematics	Semester III · CSC303
DISTRIBUTED-COMPUTING-AND-DISTRIBUTED-COMPUTING-LAB	Distributed Computing · Distributed Computing Lab	Semester VIII · CSC802, CSL802
ELECTRONIC-CIRCUITS-AND-COMMUNICATION-FUNDAMENTALS-AND-BASIC-ELECTRONICS-LAB	Electronic Circuits and Communication Fundamentals · Basic Electronics Lab	Semester III · CSC304, CSL302
FINANCE-MANAGEMENT	Finance Management	Semester VIII · ILO8022
FLAPPY-BIRD-USING-PYGAME	A Python + Pygame recreation of Flappy Bird, engineered with precise collision physics, optimized game loop timing, and arcade-style responsiveness	Technical work · HMI Lab & Neural Game Development

HADOOP	HADOOP - Matrix Multiplication	Technical work · Full Stack Engineering & Distributed Computing
HANGMAN-GAME-IN-DJANGO-PYTHON	A Hangman game web application developed using Django	Technical work · Full Stack Engineering & Distributed Computing
HANGMAN-IN-RUBY	A Ruby terminal-based Hangman game showcasing object-oriented design, modular architecture, and dynamic game state management	Technical work · Ruby Language & Algorithmic Foundations
HANGMAN-WORD-GAME	A classic graphical Hangman game implemented using Java Applets and AWT/Swing components	Semester III · Project
HUMAN-MACHINE-INTERACTION-AND-HUMAN-MACHINE-INTERACTION-LAB	Human Machine Interaction · Human Machine Interaction Lab	Semester VIII · CSC801, CSL801
JAVASCRIPT-FRAMEWORKS	Comparative Implementation of Todo Applications Across JavaScript Frameworks	Technical work · Full Stack Engineering & Distributed Computing
JULIA	50-Day Julia Programming Challenge by Amey Thakur and Mega Satish, focused on computational modeling, high-performance computing, and machine learning	Technical work · Polyglot Mastery & Logic Synthesis
KAGGLE-COURSES	A comprehensive archive of completed Kaggle Data Science, Machine Learning, and Data Manipulation courses, documenting structured Python exercises and certifications	Technical work · Machine Learning & Predictive Analytics
LUNAR-DESIGN-STUDIO	Lunar Design Studio is a professionally designed architecture portfolio project developed for Architect Mugdha Thakur	Technical work · Creative Web Design & Interface Discovery
MACHINE-LEARNING	Machine Learning	Semester VI · CSDLO6021
MANAGEMENT-INFORMATION-SYSTEM	Management Information System	Semester VII · ILO7013
MATH-SPRINT-GAME	A fast-paced educational web game where players race against time to solve math equations, testing speed and accuracy	Technical work · HMI Lab & Neural Game Development

MICROPROCESSOR-AND-MICROPROCESSOR-LAB	Microprocessor · Microprocessor Lab	Semester V · CSC501, CSL501
MOBILE-COMMUNICATION-AND-COMPUTING-AND-MOBILE-APPLICATION-DEVELOPMENT-LAB	Mobile Communication and Computing · Mobile Application Development Lab	Semester VII · CSC702, CSL702
MULTIMEDIA-SYSTEM	Multimedia System	Semester V · CSDLO5011
NATURAL-LANGUAGE-PROCESSING-AND-COMPUTATIONAL-LAB-II	Natural Language Processing · Computational Lab - II	Semester VIII · DLO8012, CSL804
ONLINE-CHESS-GAME	A real-time multiplayer chess application featuring instant move synchronization, in-game chat, and AI integration, designed with a focus on human-computer interaction principles	Semester VIII · Project
OOPM-JAVA-LAB	OOPM (Java) Lab	Semester III · CSL304
OPEN-SOURCE-TECH-LAB	Open Source Tech Lab	Semester IV · CSL405
OPERATING-SYSTEM-AND-OPERATING-SYSTEM-LAB	Operating System · Operating System Lab	Semester IV · CSC405, CSL404
OPTIMIZING-STOCK-TRADING-STRATEGY-WITH-K-MEANS-CLUSTERING	Big Data Analytics [BDA] Mini Project	Semester VII · Project
OPTIMIZING-STOCK-TRADING-STRATEGY-WITH-REINFORCEMENT-LEARNING	Machine Learning Project to Optimize Stock Trading Strategies Using Reinforcement Learning	Technical work · Machine Learning & Predictive Analytics
PONG-GAME	A modern Python + Pygame reconstruction of the original 1972 Pong, built with accurate collision physics and performance-focused game loops	Technical work · HMI Lab & Neural Game Development
PYTHON-CRASH-COURSE	IIT Ropar, Diginique Techlabs: Python Crash Course in Data Science, Machine Learning, and Artificial Intelligence	Technical work · Polyglot Mastery & Logic Synthesis
PYTHON-SHORTS	A curated collection of 100+ Python programs, algorithms, and data structure implementations for problem solving and computational practice	Technical work · Polyglot Mastery & Logic Synthesis

QUADTREE-VISUALIZER	A high-performance interactive simulation visualizing the efficiency of the QuadTree data structure in spatial partitioning and collision detection, built with Next.js and HTML5 Canvas	Semester VIII · Project
R	30-Day R Programming Challenge by Amey Thakur and Mega Satish, focused on statistical computing, data visualization, and applied data science	Technical work · Polyglot Mastery & Logic Synthesis
RAILSFRIENDS	A Ruby on Rails Friends List application showcasing MVC architecture, RESTful CRUD operations, Active Record ORM, and SQLite3 database integration	Technical work · Ruby Language & Algorithmic Foundations
REACT-TODO-APP	A Todo application developed using React	Technical work · Full Stack Engineering & Distributed Computing
RESEARCHGATE	A technical companion repository archiving code implementations and research artifacts shared on ResearchGate	Technical work · Artificial Intelligence & Neural Research
ROCK-PAPER-SCISSORS	A classic Rock Paper Scissors game implemented as a responsive web application, featuring interactive gameplay and score tracking	Technical work · HMI Lab & Neural Game Development
RUBY	30-Day Ruby Programming Challenge by Amey Thakur and Mega Satish, focused on mastering Ruby fundamentals, object-oriented programming, Ruby on Rails development, and full-stack web engineering with cloud deployment	Technical work · Ruby Language & Algorithmic Foundations
RUBY-ON-RAILS-FRIENDSAPP	Friends App Using Ruby on Rails with PostgreSQL database deployed on Heroku. [Heroku Render]	Technical work · Ruby Language & Algorithmic Foundations
SEARCH-SPACE-EXPLORE-EXTENT	It is a simple Web Design in HTML and CSS. [My first Web Design]	Technical work · Creative Web Design & Interface Discovery
SENTIMENT-ANALYZER	An NLP engine utilizing rule-based linguistic patterns for precise sentiment quantification	Technical work · Artificial Intelligence & Neural Research
SIMPLE-AND-COMPOUND-INTEREST-CALCULATOR	A Shell Script program designed to calculate simple and compound interest using user-provided inputs	Semester IV · Project

SOFTWARE-ENGINEERING-AND-SOFTWARE-ENGINEERING-LAB	Software Engineering · Software Engineering Lab	Semester VI · CSC601, CSL601
SYSTEM-PROGRAMMING-AND-COMPILER-CONSTRUCTION-AND-SYSTEM-SOFTWARE-LAB	System Programming and Compiler Construction · System Software Lab	Semester VI · CSC602, CSL602
TEXT-SUMMARIZER	Machine Learning Project to Compare and Evaluate Text Summarization Algorithms Using SpaCy, NLTK, Gensim, and Sumy	Semester VIII · Project
THE-MATH-GAME	A simple, interactive multiplication game designed to test and improve arithmetic skills through rapid-fire problem solving	Technical work · HMI Lab & Neural Game Development
THEORY-OF-COMPUTER-SCIENCE	Theory of Computer Science	Semester V · CSC504
TIC-TAC-TOE-ANGULAR-FRAMEWORK	A Tic-Tac-Toe game application developed using the Angular framework	Technical work · Full Stack Engineering & Distributed Computing
TIC-TAC-TOE-IN-RUBY	A Ruby terminal-based Tic Tac Toe game showcasing object-oriented programming (OOP), modular design, and structured game state management	Technical work · Ruby Language & Algorithmic Foundations
TSF-SUPERVISED-MACHINE-LEARNING	Task: To predict the percentage of a student based on the number of study hours	Technical work · Professional Machine Learning Missions (TSF)
TSF-UNSUPERVISED-MACHINE-LEARNING	Task: From the given 'Iris' dataset, predict the optimum number of clusters and represent it visually	Technical work · Professional Machine Learning Missions (TSF)
WEB-DESIGNING-LAB	Web Design Lab	Semester V · CSL504
WHITE-BOX-CARTOONIZATION	AI-powered web app that transforms photos into cartoon-style images using deep learning (GAN + U-Net architecture)	Semester VI · Project

Complete Directory: M.Eng.

Every repository in this document, A to Z. Each name is a link.

Repository	Subject or project	Where it sits
ADAPTIVE-CRUISE-CONTROL	Adaptive Cruise Control (ACC) with Arduino, MATLAB, and Simulink · Real-time speed regulation and proximity-based safety system	Semester II · Project
COMPUTATIONAL-INTELLIGENCE	Computational Intelligence	Semester IV · ELEC 8330
COMPUTATIONAL-METHODS-AND-MODELING-FOR-ENGINEERING-APPLICATIONS	Computational Methods and Modeling	Semester II · GENG 8030
COMPUTER-NETWORKS	Computer Networks	Semester III · ELEC 8560
DIGITAL-COMMUNICATIONS	Digital Communications	Semester IV · ELEC 8900
ENGINEERING-INTERNATIONAL-STUDENT-ADVISING	Engineering International Student Advising (ENG-ISA), University of Windsor · Academic, immigration, and student success resources for international engineering graduate students	Student resource
ENGINEERING-MATHEMATICS	Engineering Mathematics	Semester I · GENG 8010
ENGINEERING-PROJECT-MANAGEMENT	Engineering Project Management	Semester II · GENG 8020
ENGINEERING-TECHNICAL-COMMUNICATIONS	Engineering Technical Communications	Semester I · GENG 8000
INTERNATIONAL-STUDENT-CENTRE	International Student Centre (ISC), University of Windsor · Orientation Resources & Student Support Hub	Student resource
MACHINE--LEARNING	Machine Learning	Semester III · ELEC 8900
VIP-CSL-FALL-2023	VIP - Community Service Learning	Semester III · VIP-CSL
WRITING-SUPPORT	Writing Support Desk (WSD), University of Windsor · Academic writing, citation standards (APA, MLA, IEEE, Chicago), and academic integrity resources	Student resource

ZERO-SHOT-VIDEO-GENERATION	Zero-Shot Video Generation, A neural video synthesis studio powered by latent diffusion and cross-frame attention	Semester III · Project

What I Would Tell You

Six things I learned building this, offered for whatever they are worth.

1 **Keep one repository per subject.**

Not one per assignment and not one for the whole degree. A subject is the unit you will actually search for later.

2 **Write the report in the same week.**

I wrote every laboratory report while the work was still in front of me. Reconstructing one a month later takes longer and it comes out thinner.

3 **Record it while it still runs.**

Environments rot. Several of my demonstrations exist only because I recorded the session on the day, and the code that produced them will not build now.

4 **Take the notes you would want in five years.**

Not the ones that get you through the examination. Those two sets of notes look different, and only one of them is worth keeping.

5 **Publish it.**

A private archive rots quietly. Making mine public is what forced it into a state someone else could follow, and that discipline is most of the value.

6 **Index the whole thing.**

Material nobody can navigate is a folder, not an archive. The index is the part that turns four years of files into something usable.

> **Tip**
> If you take one thing from this: the work is only worth as much as your ability to find it again.

Explore Further

Where the rest of the work lives.

Related repositories

COMPUTER-ENGINEERING	The B.E. index this document is drawn from
MENG-COMPUTER-ENGINEERING	The M.Eng. index this document is drawn from
ACHIEVEMENTS	Certifications, course completions and awards
RESEARCHGATE	Code and artefacts accompanying published work
YouTube playlist	Engineering Projects, video demonstrations

Elsewhere

GitHub	github.com/Amey-Thakur
Amey's Arc	amey-thakur.github.io
ORCID	0000-0001-5644-1575
Google Scholar	Publications and citations
ResearchGate	researchgate.net/profile/Amey-Thakur
arXiv	arxiv.org/a/thakur_a_3
Kaggle	kaggle.com/ameythakur20
Hugging Face	huggingface.co/ameythakur
LinkedIn	linkedin.com/in/amey-thakur
YouTube	Engineering project demonstrations

A Closing Note

The repositories are the work. This is the index to them, written so someone arriving without context can find the part they need.

> **Important**
> Repositories change. Where this document and a repository disagree, the repository is right.

This is my own archive. It is not a course, a curriculum, an accredited programme, or an official publication of the University of Mumbai, Terna Engineering College, or the University of Windsor. The institutional marks appear only to place the work in context.

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